7+1: provenance

Sources, derivations, corrections and the verification record of the notebook 7+1.nb and of its documentation

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
Generated by tools/make_docs.py on 2026-10-26

Abstract

This record states, for every object of 7+1.nb, where it comes from (the task statement, the notebook 4+4.nb, or the literature), the equation that defines it, how it was derived, and how it was checked: the full list of the 202 checks and their results, the independent Python cross-checks, and the hashes of the files. Sections 1–3 concern the first stage (sections 0–6 of the notebook: octonions and Clifford algebras), section 6 the second stage (sections 7–10: the Dirac equation and the Kohn–Sham theories of the field dirac16complex).

Contents

1. Sources **2**
1.1 The task statement . . . 2
1.2 The notebook 4+4.nb . . . 2
1.3 Literature . . . 3

2. Derivations: how each new object was obtained **3**
2.1 Octonion matrices τ^O . . . 3
2.2 Octonion vielbein and multiplication constants . . . 3
2.3 Cayley–Dickson form . . . 3
2.4 Euclidean mirror T^O . . . 3
2.5 Real 7+1 generators $\Gamma^{(1,7)}$ . . . 4
2.6 Complexified generators $\Gamma^{(7,1)}$ . . . 4
2.7 The impossibility theorem . . . 4
2.8 Real 32×32 generators . . . 4
2.9 Examples . . . 4
2.10 Classification of the nine signatures . . . 5

3. Corrections to the task statement (with the evidence in 7+1.nb) **5**

4. Verification record **5**
4.1 The Mathematica run . . . 5
4.2 All checks of 7+1.nb and their results . . . 5
4.3 Independent Python cross-checks . . . 15
4.4 Hashes of the files used to build 7+1.nb and its documentation . . . 20

5. Process and environment **21**

6. The second stage: sections 7–10 of 7+1.nb **22**
6.1 The task . . . 22
6.2 The first part of the notebook is unchanged . . . 22
6.3 Derivations and decisions . . . 22
6.4 Literature used in the second stage . . . 24
6.5 Verification of the second stage . . . 24

7. Authorship **24**

1. Sources

1.1 The task statement

The task asked for a new Mathematica notebook 7+1.nb in `D:\Developer\github\7+1` defining a new realization of the octonions and a new representation of the Clifford algebra of an 8-dimensional vector space that mirror, as far as possible, the split octonions and the pseudo-Clifford algebra of 4+4.nb; real numbers and the real 8×8 and 16×16 matrices of 4+4.nb were to be used if possible, otherwise their complexification; examples were to check all work; notation was to be changed and documented where confusing. It fixed the coordinates

$$\begin{aligned}
&\text{4+4.nb:}\quad X=(x_0,\dots,x_7),\ x_0=\text{hidden space-like},\ x_1,x_2,x_3=\text{3-space},\\
&\qquad\qquad x_4=\text{time},\ x_5,x_6,x_7=\text{auxiliary time-like},\\
&\text{7+1.nb:}\quad X=(x_0,\dots,x_7),\ x_0=\text{time (octonion unit)},\ x_1,x_2,x_3=\text{3-space},\\
&\qquad\qquad x_4,\dots,x_7=\text{auxiliary space-like},
\end{aligned}$$

and gave two definitions. [1A] the octonions: $e_0=1$, $e_i^2=-1$, Fano plane, Cayley–Dickson doubling $\mathbb{O}=\mathbb{H}\oplus\mathbb{H}$, conjugation $\bar{x}=x_0-\sum_i x_i e_i$ with $\overline{xy}=\bar{y}\bar{x}$, $N(x)=x\bar{x}=\sum_A x_A^2$, $N(xy)=N(x)N(y)$, $x^{-1}=\bar{x}/N(x)$, alternativity, power-associativity. [1B] the Clifford algebra $Cl(V,Q)=\mathcal{T}(V)/\mathcal{I}_Q(V)$ with $v^2=Q(v)1$, $vw+wv=2B(v,w)1$, $B(v,w)=\frac{1}{2}(Q(v+w)-Q(v)-Q(w))$, dimension $2^8=256$, a classification table for $p+q=8$, and $Cl_8(\mathbb{C})\cong\mathbb{C}(16)$. Section 4 lists the statements of [1A] and [1B] that had to be corrected.

1.2 The notebook 4+4.nb

File `4+4.nb`, 9348656 bytes, written by Wolfram 15.0 (header `CreatedBy='Wolfram 15.0'`), SHA-256 `740023233eaa0cd24ca0e2e7010094671b1d314b698114c08db94050d3e0aed5`. It was read with the parser `tools/nbparse.py` (806 cells: 366 input, 366 output, 49 text). Every object used by 7+1.nb was reproduced with the formula of the cell given here, and each reproduction is checked in 7+1.nb section 2:

object	4+4.nb cell	definition in 4+4.nb
$\eta^{(4,4)}$	In[9]	$\mathrm{diag}(1_4,-1_4)$ (`η4488`)
σ	In[18]	$\left(\begin{smallmatrix}0 & 1_4\\ 1_4 & 0\end{smallmatrix}\right)$ (`σ=σ88`)
s_h, t_h	In[29], In[35], In[36]	$(s_h)_{pq}=\epsilon_{hpq4}-(\delta_{p4}\delta_{qh}-\delta_{ph}\delta_{q4})$, $(t_h)_{pq}=\epsilon_{hpq4}+(\dots)$
ζ_A	In[60]–In[63]	$\zeta_0=1_8$, $\zeta_{7-h}=\left(\begin{smallmatrix}0 & t_h\\ -t_h & 0\end{smallmatrix}\right)$, $\zeta_h=\left(\begin{smallmatrix}0 & s_h\\ s_h & 0\end{smallmatrix}\right)$, $\zeta_7=\zeta_1\cdots\zeta_6$
relations of ζ	In[64], In[65], In[67]	$\sigma=\zeta_1\zeta_2\zeta_3$; $\frac{1}{2}\{\zeta_A,\zeta_B\}=-\eta^{(4,4)}_{AB}1_8$; $-\eta^{(4,4)}_{AA}\zeta_A=\zeta_A^{\mathsf{T}}$
τ_A	In[70]	$\tau_0=1_8$, $\tau_A=\zeta_A$
$\bar{\tau}_A$	In[74]	$\bar{\tau}_0=1_8$, $\bar{\tau}_A=\sigma\tau_A^{\mathsf{T}}\sigma$ (symbolized tau-bar)
relations of τ	In[77], In[87], In[89]	$\frac{1}{2}(\tau_A\bar{\tau}_B+\tau_B\bar{\tau}_A)=\eta^{(4,4)}_{AB}1_8$; $\tau_1\cdots\tau_7=1_8$; $\sigma=\tau_4\tau_5\tau_6\tau_7$
T_A	In[98]	$T_A=\left(\begin{smallmatrix}0 & \bar{\tau}_A\\ \tau_A & 0\end{smallmatrix}\right)$ (`T16^A[A1]`)
T_8, σ_{16}	In[99], In[97], In[106]	$T_8=T_0\cdots T_7$; $\sigma_{16}=T_0T_1T_2T_3=\mathrm{diag}(-\sigma,\sigma)$
Clifford relation	In[110]	$\frac{1}{2}\{T_A,T_B\}=\eta^{(4,4)}_{AB}1_{16}$
P_L, P_R	In[112]–In[114]	$P_L=\frac{1}{2}(1_{16}-T_8)$, $P_R=\frac{1}{2}(1_{16}+T_8)$
U_h	In[257], In[259]	eigenvectors of σ divided by $\sqrt{2}$
u	In[264], In[266]	`hUSE = 8`, $u=U_8=(1,0,0,0,1,0,0,0)/\sqrt{2}$
g	In[283]	$g_{AB}=\eta^{(4,4)}$
E_{44}, E_{44}^{-1}	In[294]–In[296]	$(E_{44})_{Ba}=(U_h^{\mathsf{T}}\sigma\bar{\tau}_B)_a$; $E_{44}^{-1}=(U_h^{\mathsf{T}}\sigma U_h)(\eta^{(4,4)}_{BB}\tau_B U_h)_B$
ε_B	In[313]	`EA = Array[eA, 8]`
split table	In[325], Out[325]	$\varepsilon_B\varepsilon_C=\sum_D(g_{BB}E_{44}\tau_{B-1}E_{44}^{-1})_{DC}\varepsilon_D$ (`gridx`)

1.3 Literature

- J. C. Baez, *The Octonions*, Bull. Amer. Math. Soc. 39 (2002) 145–205, arXiv:math/0105155, section 2.2 (the page https://math.ucr.edu/home/baez/octonions/node5.html, which the task statement itself cites, was read on 2026-10-26): the Cayley–Dickson product $(a,b)(c,d) = (ac - db^*,\ a^*d + cb)$, where Baez writes x^* for the conjugate $\bar{x}$ and the conjugate of a pair is $(a,b)^* = (a^*, -b)$. 7+1.nb proves that its octonions satisfy this product with $x = a + \ell b$, $\ell = e_4$.
- H. B. Lawson and M.-L. Michelsohn, *Spin Geometry*, Princeton (1989): classification of real Clifford algebras, $Cl(p+1,q+1) \cong Cl(p,q) \otimes \mathbb{R}(2)$. 7+1.nb does not rely on it: it computes the classification directly (section 2.10 below).
- Standard facts used in proofs: Schur's lemma (the intertwiners of an irreducible module form a division algebra; over $\mathbb{C}$, the scalars); $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$ has a unique irreducible module $\mathbb{C}^{16}$; Hurwitz's theorem (a unital composition algebra of dimension 8 is the octonions if its norm is positive definite and the split octonions if the norm is indefinite); Degen's eight-square identity.

2. Derivations: how each new object was obtained

2.1 Octonion matrices τ^O

Origin: the split matrices of 4+4.nb and the observation (checked) that $\sigma = \tau_1\tau_2\tau_3 = \tau_4\tau_5\tau_6\tau_7$ commutes with τ_1, τ_2, τ_3, anticommutes with $\tau_4, \ldots, \tau_7$ and squares to 1_8. Hence

$$\tau_0^O = 1_8, \quad \tau_A^O = \tau_A \ (A \le 3), \quad \tau_k^O = \tau_k\sigma \ (k \ge 4), \qquad (\tau_k\sigma)^2 = \tau_k\sigma\tau_k\sigma = -\tau_k^2\sigma^2 = -1_8,$$
$$\tau_h^O\tau_k^O = \tau_h\tau_k\sigma = -\tau_k\tau_h\sigma = -\tau_k\sigma\tau_h = -\tau_k^O\tau_h^O, \qquad \tau_k^O\tau_l^O = \tau_k\sigma\tau_l\sigma = -\tau_k\tau_l = \tau_l\tau_k = -\tau_l^O\tau_k^O \quad (k \ne l \ge 4).$$

This is the real counterpart of the complex substitution $\tau_k \to i\tau_k$ that turns split octonions into octonions over $\mathbb{C}$.

2.2 Octonion vielbein and multiplication constants

Origin: In[294] and In[325] of 4+4.nb with the split spinor metric σ replaced by the Euclidean $g^O = 1_8$ and the same unit spinor u:

$$(E^O)_{Ba} = \left(u^{\mathsf{T}} g^O \bar{\tau}_B^O\right)_a, \qquad L_A^O = g_{AA}^O E^O \tau_A^O (E^O)^{-1}, \qquad e_A e_K = \sum_L (L_A^O)_{LK} e_L.$$

Why the result is the octonion algebra: $(E^O)^{-1}e_0 = u$ makes e_0 a two-sided unit, and $\tau^O(x)\bar{\tau}^O(x) = N(x)1_8$ together with the orthogonality of E^O gives $N(xy) = N(x)N(y)$. A unital composition algebra of dimension 8 with a positive definite norm is the octonions (Hurwitz). The notebook checks all the laws directly.

2.3 Cayley–Dickson form

Origin: Baez 2002, section 2.2. With $\ell = e_4$, $x = a + \ell b$, $a = (x_0, \ldots, x_3)$, $b = (x_4, \ldots, x_7)$:

$$(a,b)(c,d) = \left(ac + \gamma\, d\bar{b},\ \bar{a}d + cb\right), \qquad \gamma = -1 \text{ (7+1.nb)}, \qquad \gamma = +1 \text{ (4+4.nb)}.$$

2.4 Euclidean mirror T^O

Origin: In[98] of 4+4.nb with $\tau \to \tau^O$: $T_A^O = \begin{pmatrix} 0 & \bar{\tau}_A^O \\ \tau_A^O & 0 \end{pmatrix}$, so that $\{T_A^O, T_B^O\} = \begin{pmatrix} \bar{\tau}_A^O\tau_B^O + \bar{\tau}_B^O\tau_A^O & 0 \\ 0 & \tau_A^O\bar{\tau}_B^O + \tau_B^O\bar{\tau}_A^O \end{pmatrix} = 2\delta_{AB}1_{16}$. In terms of 4+4.nb: $T_A^O = T_A$ $(A \le 3)$, $T_k^O = T_k S_{44}$ $(k \ge 4)$, $S_{44} = T_4T_5T_6T_7 = \mathrm{diag}(\sigma, \sigma)$.

2.5 Real 7+1 generators $\Gamma^{(1,7)}$

Origin: the standard construction of a Lorentzian Clifford algebra from a Euclidean one, applied to T^O:

$$\Gamma_0^{(1,7)} = T_0^O, \qquad \Gamma_i^{(1,7)} = T_0^O T_i^O, \qquad (T_0^O T_i^O)^2 = -(T_0^O)^2 (T_i^O)^2 = -1_{16},$$
$$\{T_0^O, T_0^O T_i^O\} = T_i^O + T_0^O T_i^O T_0^O = T_i^O - T_i^O = 0, \qquad \{T_0^O T_i^O, T_0^O T_j^O\} = -\{T_i^O, T_j^O\} = -2\delta_{ij} 1_{16}.$$

Hence $\{\Gamma_A^{(1,7)}, \Gamma_B^{(1,7)}\} = 2\eta_{AB}^{(1,7)} 1_{16}$, and $\Gamma^{(1,7)}(x)^2 = \mathrm{Re}(xx) 1_{16}$.

2.6 Complexified generators $\Gamma^{(7,1)}$

Origin: the task's instruction to use the complexification of the 4+4 matrices if real numbers fail. Only the time block is multiplied by i: $\tau_0^C = \bar\tau_0^C = i 1_8$, $\tau_k^C = \tau_k^O$; then $\Gamma_A^{(7,1)} = \left(\begin{smallmatrix} 0 & \bar\tau_A^C \\ \tau_A^C & 0 \end{smallmatrix}\right)$, $(\Gamma_0^{(7,1)})^2 = i^2\, 1_{16} = -1_{16}$, and $\Gamma_A^{(7,1)} = S(i\Gamma_A^{(1,7)})S^{-1}$ with $S = (1_{16} - i\Gamma_0^{(1,7)})/\sqrt{2}$.

2.7 The impossibility theorem

Origin: Schur's lemma and the uniqueness of the irreducible module of $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$. The intertwiners M with $M\overline{\Gamma_A^{(7,1)}} = \Gamma_A^{(7,1)} M$ are computed in 7+1.nb with `NullSpace`; they form the line through $M_0 = \Gamma_8^{(1,7)} = \left(\begin{smallmatrix} 0 & -1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, and $M_0\overline{M_0} = -1_{16}$. A real realization $R_A = T\Gamma_A^{(7,1)}T^{-1}$ would give an intertwiner $M = T^{-1}\overline{T}$ with $M\overline{M} = 1_{16}$, impossible because $M = cM_0$ gives $M\overline{M} = -|c|^2 1_{16}$.

2.8 Real 32×32 generators

Origin: realification. $\widehat\Gamma_A^{(7,1)} = \varepsilon \otimes \Gamma_A^{(1,7)}$ with $\varepsilon^2 = -1_2$ gives $\{\widehat\Gamma_A^{(7,1)}, \widehat\Gamma_B^{(7,1)}\} = \varepsilon^2 \otimes 2\eta_{AB}^{(1,7)} 1_{16} = 2\eta_{AB}^{(7,1)} 1_{32}$. Commutant: $Q_i = \varepsilon \otimes 1_{16}$, $Q_j = \sigma_1 \otimes \Gamma_8^{(1,7)}$, $Q_k = \sigma_3 \otimes \Gamma_8^{(1,7)}$, found from $\varepsilon\sigma_1 = -\sigma_1\varepsilon$, $\varepsilon\sigma_3 = -\sigma_3\varepsilon$ and $\Gamma_8^{(1,7)}\Gamma_A^{(1,7)} = -\Gamma_A^{(1,7)}\Gamma_8^{(1,7)}$.

2.9 Examples

so(1,7): $\Sigma_{AB} = \frac{1}{4}[\Gamma_A^{(1,7)}, \Gamma_B^{(1,7)}]$; the commutation relations follow from the Clifford relation. Boost and rotation: $(\Gamma_0^{(1,7)}\Gamma_1^{(1,7)})^2 = 1_{16}$, $(\Gamma_4^{(1,7)}\Gamma_5^{(1,7)})^2 = -1_{16}$, so $e^{\beta\Sigma_{01}} = \cosh\frac{\beta}{2} + \sinh\frac{\beta}{2}\Gamma_0^{(1,7)}\Gamma_1^{(1,7)}$ and $e^{\theta\Sigma_{45}} = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}\Gamma_4^{(1,7)}\Gamma_5^{(1,7)}$. Dirac operator: $(\Gamma^A\partial_A)^2 = \frac{1}{2}\{\Gamma^A, \Gamma^B\}\partial_A\partial_B = \eta_{(1,7)}^{AB}\partial_A\partial_B$. Real amplitudes: $p = (3,2,2,0,0,0,0,0)$, $p^{\mathsf T}\eta^{(1,7)}p = 1$, $\not{p}^{\,2} = 1_{16}$; the real matrix $\not{p}$ has a real 8-dimensional eigenspace for the eigenvalue $+1$.

Real spinor fields. If J anticommutes with every $\Gamma_A^{(1,7)}$, then $J\Gamma_8^{(1,7)}$ commutes with every $\Gamma_A^{(1,7)}$ (because $\Gamma_8^{(1,7)}$ anticommutes with each of them), so by irreducibility $J\Gamma_8^{(1,7)} = c\, 1_{16}$ and $J = -c\,\Gamma_8^{(1,7)}$ (as $(\Gamma_8^{(1,7)})^{-1} = -\Gamma_8^{(1,7)}$), whose square is $-c^2\, 1_{16}$. Hence the real operator $\Gamma^A\partial_A - mJ$ has the square $\partial_0^2 - \cdots - \partial_7^2 - c^2m^2$ (a tachyon), and a real equation $(G^A\partial_A - m)\Psi = 0$ implies the Klein–Gordon equation only if the real G_A obey the $\eta^{(7,1)}$ relation, impossible for 16×16 matrices (2.7). Massless: for $k = (1,1,0,\ldots,0)$, $\not{k}^{\,2} = k^{\mathsf T}\eta^{(1,7)}k\, 1_{16} = 0$ and $\Gamma^A\partial_A[w\cos(k\cdot x)] = -\sin(k\cdot x)\ \not{k}\, w$, which vanishes for the 8 real solutions of $\not{k}\, w = 0$. Massive, with the real 32×32 matrices: $P = \sum_A p^A\widehat\Gamma_A^{(7,1)}$, $P^2 = -m^2\, 1_{32}$, and with $\phi = p\cdot x$, $\widehat\Gamma^A = \eta_{(7,1)}^{AA}\widehat\Gamma_A^{(7,1)}$:

$$\begin{aligned}\widehat\Gamma^A\partial_A\Big(w\cos\phi + \tfrac{1}{m}Pw\sin\phi\Big) &= -P\Big(-w\sin\phi + \tfrac{1}{m}Pw\cos\phi\Big)\\ &= Pw\sin\phi + m\,w\cos\phi = m\Big(w\cos\phi + \tfrac{1}{m}Pw\sin\phi\Big),\end{aligned}$$

using $\partial_A\phi = (\eta^{(1,7)}p)_A$ and $\eta_{(7,1)}^{AA}\eta_{AA}^{(1,7)} = -1$.

2.10 Classification of the nine signatures

For each p, complex generators $G_A = T_A^O$ $(A < p)$, $G_A = iT_A^O$ $(A \geq p)$ satisfy $\{G_A, G_B\} = \pm 2\delta_{AB} 1_{16}$ with p plus signs. The indicator sign of $M\overline{M}$ ($M\overline{G_A} = G_A M$) is $+1$ for $p \in \{8,5,4,1,0\}$ ($\mathbb{R}(16)$) and -1 for $p \in \{7,6,3,2\}$ ($\mathbb{H}(8)$). An independent numerical method (`tools/classify_clifford.py`: commutant of the realified 32-dimensional representation, whose traceless part has a negative definite trace form exactly in the quaternionic case) gives the same table. Explicit real generators: T^O (8,0), $\Gamma_0^{(5,3)} = T_0$, $\Gamma_i^{(5,3)} = T_0 T_i$ (5,3) with $\eta^{(5,3)} = \mathrm{diag}(1,-1,-1,-1,1,1,1,1)$, T (4,4), $\Gamma^{(1,7)}$ (1,7), $\Gamma_A^{(0,8)} = T_8^O T_A^O$ (0,8).

3. Corrections to the task statement (with the evidence in 7+1.nb)

statement in the task	correction	evidence (check in 7+1.nb)
[1A]: $(a,b)(c,d) = (ac - \bar{d}b,\ \bar{a}d + cb)$ defines the octonions	not alternative, norm not multiplicative; correct form $(ac - d\bar{b},\ \bar{a}d + cb)$ (Baez 2002, 2.2)	"the formula ... of [1A] is NOT alternative", "... its norm is not multiplicative", "Cayley-Dickson, γ = -1 ..."
[1A]: first formula $(ac - \bar{d}b,\ da + cb\bar{x})$	contains a stray "x"; not interpretable	—
[1B]: $1+8+28+56+72+56+28+8+1 = 256$	$\binom{8}{4} = 70$	"grades: binomial(8,k) == ..."
[1B]: $Cl(7,1) \cong \mathbb{R}(16)$	$Cl(7,1) \cong \mathbb{H}(8)$ for $v^2 = Q(v)$	"indicator(Γ71) == -1 ...", "classification: ..."
[1B]: $Cl(5,3) \cong \mathbb{C}(16)$, $Cl(3,5) \cong \mathbb{C}(16)$	$Cl(5,3) \cong \mathbb{R}(16)$, $Cl(3,5) \cong \mathbb{H}(8)$; $\mathbb{C}(16)$ has real dimension 512	"classification: ...", "$\mathbb{C}(16)$ cannot occur ..."
[1B]: metric-sign strings for (6,2), (5,3), (4,4), (2,6)	mistyped (wrong number of signs)	table of section 5.6 of 7+1.nb
split octonions are a "pseudo-normed division algebra"	they have zero divisors: $(\varepsilon_1 + \varepsilon_5)(\varepsilon_1 - \varepsilon_5) = 0$	"split: (eA[1] + eA[5])(eA[1] - eA[5]) == 0 ..."

4. Verification record

4.1 The Mathematica run

item	value
program	`C:\Program Files\Wolfram Research\Wolfram\15.0.1\WolframNB.exe`
$Version	15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)
method	`Get["D:/Developer/github/7+1/tools/run_eval.wl"]`: `NotebookEvaluate[nb, InsertResults -> True]`, then `NotebookSave`
finished	Mon 26 Oct 2026 17:48:46
evaluation time	16.1455 s
private context of the notebook	`Notebook$$38$50910`
input cells / output cells / message cells	92 / 241 / 0
checks / True	202 / 202
commutant computation (8192×1024 sparse `NullSpace`)	1.0547124 s
SHA-256 of the evaluated 7+1.nb (604743 bytes)	`438d972073e2fef9d7da6d57a541432b0fb1ea08424307f59ae2f3f7752bbfaf`

4.2 All checks of 7+1.nb and their results

Each row is one cell `check["name", test]` of the notebook, in order, grouped by the section of the notebook in which it stands; the test is the Wolfram Language expression in that cell, and the result is what Wolfram 15.0.1 returned.

2.1 σ, the self-dual and anti-self-dual blocks, ζ, τ and τbar

#	check	result
1	4+4: ζ[7] == diag(-1,-1,-1,-1,1,1,1,1)	True
2	4+4: σ == ζ[1].ζ[2].ζ[3]	True
3	4+4: (ζ[A].ζ[B] + ζ[B].ζ[A])/2 == -η4488[[A+1,B+1]] ID8 (A,B = 1..7)	True
4	4+4: -η4488[[A+1,A+1]] ζ[A] == Transpose[ζ[A]] (A = 1..7)	True
5	4+4: (τ[A].τbar[B] + τ[B].τbar[A])/2 == η4488[[A+1,B+1]] ID8	True
6	4+4: (τbar[A].τ[B] + τbar[B].τ[A])/2 == η4488[[A+1,B+1]] ID8	True
7	4+4: τ[1].τ[2]...τ[7] == ID8	True
8	4+4: σ == τ[1].τ[2].τ[3] == τ[4].τ[5].τ[6].τ[7]	True
9	4+4: σ commutes with τ[1..3] and anticommutes with τ[4..7]; σ.σ == ID8	True

2.2 The 4+4 Dirac matrices T16 and the chiral projectors

#	check	result
10	4+4: (T16[A].T16[B] + T16[B].T16[A])/2 == η4488[[A+1,B+1]] ID16	True
11	4+4: T16[A] real (entries 0, ±1)	True
12	4+4: T16[8] == diag(-ID8, ID8)	True
13	4+4: σ16 == diag(-σ, σ)	True
14	4+4: PL44 + PR44 == ID16, PL44.PL44 == PL44, PR44.PR44 == PR44, PL44.PR44 == 0	True

2.3 The split-octonion multiplication constants (gridx of 4+4.nb) and how they come from τ

#	check	result
15	4+4: uBasis == evecs/Sqrt[2] from Eigensystem[σ] (the basis of 4+4.nb)	True
16	4+4: unit == (1,0,0,0,1,0,0,0)/Sqrt[2], σ.unit == unit, unit.unit == 1, unit.σ.unit == 1	True
17	4+4 (In[294]): E44inv[h] == (u.σ.u) Transpose[Table[η4488[[B,B]] τ[B-1].u, {B,1,8}]] for h = 1..8	True
18	4+4: gridx44 is identical to the split-octonion table printed by 4+4.nb	True
19	4+4: LS[A] are signed permutation matrices (constants 0, ±1)	True

2.4 Split octonions: composition algebra, but not a division algebra

#	check	result
20	split: eA[1] is the identity; eA[2..4]^2 == -eA[1]; eA[5..8]^2 == +eA[1]	True
21	split: x x̄ == N(x) eA[1] with N = η4488 (symbolic)	True
22	split: N(xy) == N(x) N(y) (symbolic)	True
23	split: alternative x(xy) == (xx)y and (yx)x == y(xx) (symbolic)	True
24	split: (eA[1] + eA[5])(eA[1] - eA[5]) == 0 with both factors nonzero: NOT a division algebra	True

3.1 From the split matrices τ to the octonionic matrices τO

#	check	result
25	τO[A] (A = 1..7): real, antisymmetric, τO[A].τO[A] == -ID8	True
26	τO[A] anticommute: τO[A].τO[B] == -τO[B].τO[A] (A != B, A,B = 1..7)	True
27	mirror of 4+4: (τO[A].τObar[B] + τO[B].τObar[A])/2 == gO[[A+1,B+1]] ID8	True
28	mirror of 4+4: (τObar[A].τO[B] + τObar[B].τO[A])/2 == gO[[A+1,B+1]] ID8	True

#	check	result
29	mirror of 4+4: τO[1].τO[2]...τO[7] == ID8	True
30	mirror of 4+4: τObar[A] == gO.Transpose[τO[A]].gO, as τbar[A] == σ.Transpose[τ[A]].σ	True
31	τO[4..7] are products of three split τ: τO[4] == τ[5].τ[6].τ[7], τO[5] == -τ[4].τ[6].τ[7], τO[6] == τ[4].τ[5].τ[7], τO[7] == -τ[4].τ[5].τ[6]	True

3.2 The octonion vielbein (the same unit spinor u as in 4+4.nb)

#	check	result
32	σ.unit == unit, hence τO[A].unit == τ[A].unit (A = 0..7)	True
33	EO is orthogonal: EOinv == Transpose[EO]	True
34	mirror of 4+4.nb In[294]: EOinv == (u.gO.u) Transpose[Table[gO[[B,B]] τO[B-1].u, {B,1,8}]]	True
35	EO == η4488.E44[hUSE]: the two vielbeins differ only by the signs of rows 5..8	True

3.3 The octonion multiplication constants: gridO, the mirror of gridx

#	check	result
36	gridO equals the octonion table computed independently in Python	True
37	LO[A] are signed permutation matrices (all multiplication constants are 0, ±1)	True
38	oMul reproduces gridO on all 64 pairs of basis units	True

3.4 The properties listed in [1A], verified

#	check	result
39	e[0] is a two-sided identity	True
40	e[i] e[i] == -e[0] (i = 1..7)	True
41	e[i] e[j] == -e[j] e[i] (i != j, i,j = 1..7): non-commutative	True
42	7 Fano lines; every pair of distinct imaginary units lies on exactly one line	True
43	each oriented line (a,b,c) is cyclic: e[a]e[b] == e[c], e[b]e[c] == e[a], e[c]e[a] == e[b]	True
44	each line spans a quaternion subalgebra: {e[0], e[a], e[b], e[c]} closed and associative	True
45	the picture's 7 lines are exactly fanoLines	True
46	left alternative: x(xy) == (xx)y (symbolic)	True
47	right alternative: (yx)x == y(xx) (symbolic)	True
48	flexible: (xy)x == x(yx) (symbolic)	True
49	associator is alternating: [x,y,z] == -[y,x,z] == -[x,z,y] (symbolic)	True
50	power associativity: x(xx) == (xx)x (symbolic)	True
51	Moufang: z(x(zy)) == ((zx)z)y, x(z(yz)) == ((xz)y)z, (zx)(yz) == (z(xy))z (x, y basis units, z symbolic)	True
52	x x̄ == x̄ x == N(x) e[0] (symbolic)	True
53	conjugation is an anti-automorphism: conj(xy) == conj(y) conj(x) (symbolic)	True
54	Re(x) == (x + x̄)/2 == x0	True
55	norm is multiplicative: N(xy) == N(x) N(y) (symbolic; Degen's eight-square identity)	True
56	inverse x^-1 = x̄/N(x): x x^-1 == x^-1 x == e[0] (symbolic)	True
57	every x satisfies x x - 2 Re(x) x + N(x) e[0] == 0, so the subalgebra generated by x is ℝ or ℂ	True
58	exactly 168 of the 512 ordered basis triples are non-associative (= 7*6*4: distinct imaginary units not on one Fano line)	True
59	example: (e[1]e[2])e[4] == e[7] but e[1](e[2]e[4]) == -e[7]	True
60	numerical example: N(x y) == N(x) N(y) == 15*6 == 90 and x x^-1 == e[0]	True
61	numerical example: x y != y x and (x y) z != x (y z)	True

3.5 Cayley–Dickson doubling: $\mathbb{O} = \mathbb{H} \oplus \ell\mathbb{H}$

#	check	result
62	$\mathbb{H}$ = span{e[0],e[1],e[2],e[3]} is a subalgebra	True
63	ℓ = e[4] doubles $\mathbb{H}$: e[4] e[k] == e[4+k] (k = 1,2,3) and e[4] e[4] == -e[0]	True
64	orientation inherited from 4+4.nb: e[1] e[2] == -e[3]	True
65	Cayley-Dickson, γ = -1: (ac - d $\bar{b}$, $\bar{a}$ d + c b) reproduces the octonion product (symbolic)	True
66	Cayley-Dickson, γ = +1: (ac + d $\bar{b}$, $\bar{a}$ d + c b) reproduces the SPLIT product of 4+4.nb (symbolic)	True
67	the formula (ac - $\bar{d}$ b, $\bar{a}$ d + c b) of [1A] is NOT alternative: x·x·y != x(xy) for x = e[0]+e[4], y = e[1]	True
68	... and its norm is not multiplicative (symbolic)	True

3.6 Octonions versus split octonions

#	check	result
69	e[A]e[B] (octonion) == s_AB e[A]e[B] (split) for all A, B = 0..7	True
70	over $\mathbb{C}$: ϕ(x *split y) == ϕ(x) *oct ϕ(y) (symbolic): split $\otimes\ \mathbb{C} \cong \mathbb{O} \otimes \mathbb{C}$	True
71	the norms correspond: N_oct(ϕ(x)) == N_split(x) (symbolic)	True

4.1 The exact mirror of T16: the Euclidean Clifford algebra Cl(8,0)

#	check	result
72	Cl(8,0): (T16O[A].T16O[B] + T16O[B].T16O[A])/2 == gO[[A+1,B+1]] ID16	True
73	S44 == diag(σ, σ)	True
74	T16O from the 4+4 matrices: T16O[A] == T16[A] (A = 0..3), T16O[k] == T16[k].S44 (k = 4..7)	True
75	T16O[8] == T16[8] == diag(-ID8, ID8): same chirality matrix and same PL44, PR44 as 4+4.nb	True

4.2 The REAL Clifford algebra of 7+1 spacetime: Γ17, Cl(1,7) ≅ $\mathbb{R}$(16)

#	check	result
76	7+1 REAL Clifford relation: (Γ17[A].Γ17[B] + Γ17[B].Γ17[A])/2 == η17[[A+1,B+1]] ID16 == -η71[[A+1,B+1]] ID16	True
77	Γ17[A] are REAL: every entry is 0 or ±1	True
78	block form: Γ17[0] == {{0,ID8},{ID8,0}}, Γ17[i] == {{τO[i],0},{0,τObar[i]}}	True
79	built from the real 4+4 matrices: Γ17[0] == T16[0], Γ17[i] == T16[0].T16[i] (i = 1,2,3), Γ17[k] == T16[0].T16[k].S44 (k = 4..7)	True
80	Γ17[0] symmetric, Γ17[i] antisymmetric, so Transpose[Γ17[A]] == Γ17[0].Γ17[A].Γ17[0] (real Dirac adjoint)	True
81	the 256 products Γ17[A1]...Γ17[Ak] (A1 < ... < Ak) are linearly independent: Cl(1,7) ≅ $\mathbb{R}$(16)	True
82	grades: binomial(8,k) == {1,8,28,56,70,56,28,8,1}, total 256 (binomial(8,4) == 70, not 72)	True
83	Tr[product] == 0 for every non-empty product, so Tr[M]/16 is the scalar part of M	True
84	irreducible over $\mathbb{R}$: only multiples of ID16 commute with all Γ17[A] (Schur; real commutant of dimension 1)	True
85	universal property: Γ17(v).Γ17(v) == (v.η17.v) ID16 for a symbolic vector v = X	True

4.3 The complexified 7+1 realization in the exact block form of 4+4.nb: Γ71

#	check	result
86	7+1 COMPLEX Clifford relation, 4+4.nb sign: (Γ71[A].Γ71[B] + Γ71[B].Γ71[A])/2 == η71[[A+1,B+1]] ID16	True
87	Γ71[0] == I T16[0] and Γ71[k] == T16O[k] (k = 1..7): only the time component is complex	True
88	S = (ID16 - I Γ17[0])/Sqrt[2] is unitary and Γ71[A] == S.(I Γ17[A]).Inverse[S]	True

4.4 Theorem: with η71 there is no real 16×16 realization, because Cl(7,1) ≅ $\mathbb{H}$(8)

#	check	result
89	Schur: the intertwiners M with M.Conjugate[Γ71[A]] == Γ71[A].M form a 1-dimensional space	True
90	... spanned by the volume element Γ17[8] == {{0,-ID8},{ID8,0}}, and Γ17[8].Γ17[8] == -ID16	True
91	indicator(Γ71) == -1: Cl(7,1) ≅ ℍ(8), NO real 16×16 realization of the η71 relation	True
92	indicator(Γ17) == +1: Cl(1,7) ≅ ℝ(16), real realization exists (Γ17)	True

4.5 A real realization of Cl(7,1) needs 32×32 matrices

#	check	result
93	REAL 32×32 realization of the η71 relation: (Γ71R32[A].Γ71R32[B] + Γ71R32[B].Γ71R32[A])/2 == η71[[A+1,B+1]] ID32	True
94	Qi, Qj, Qk commute with every Γ71R32[A]; Qi.Qi == Qj.Qj == Qk.Qk == -ID32; Qi.Qj == Qk == -Qj.Qi	True
95	the commutant of {Γ71R32[A]} has dimension exactly 4 and is spanned by ID32, Qi, Qj, Qk: it is ℍ	True

4.6 Volume elements and chirality

#	check	result
96	Γ17[8].Γ17[8] == -ID16 and Γ17[8] anticommutes with every Γ17[A]: no real chirality in 7+1	True
97	Γ71[8].Γ71[8] == -ID16 and χ71 = -I Γ71[8] == T16O[8] == T16[8] (real, diagonal)	True
98	7+1 chiral projectors in the Γ71 basis are exactly PL44, PR44 of 4+4.nb; χ71 anticommutes with every Γ71[A]	True
99	complex chiral projectors for Γ17: PL17 + PR17 == ID16, PL17.PL17 == PL17, PL17.PR17 == 0, rank 8 each	True

4.7 What the octonions do: Γ(x)² = Re(x x), and alternativity

#	check	result
100	Re(x x) == X.η17.X: the octonion square produces the 7+1 interval (symbolic)	True
101	octonion basis: Γ17e[0] == {{0,ID8},{ID8,0}} and Γ17e[i] == {{LO[i],0},{0,-LO[i]}}	True
102	alternativity in action: xImag(xImag ψ) == -|xImag|^2 ψ for the imaginary part xImag of x (symbolic)	True
103	T16O(x)^2 == N(x) ID16 (octonion norm) and T16(x)^2 == (X.η4488.X) ID16 (4+4.nb)	True

5.1 The Lorentz algebra so(1,7)

#	check	result
104	28 real generators Σ17[A,B] (A < B), linearly independent	True
105	[Σ17[A,B], Γ17[K]] == η17[[B+1,K+1]] Γ17[A] - η17[[A+1,K+1]] Γ17[B]	True
106	so(1,7): [Σ_AB, Σ_KL] == η_BK Σ_AL - η_AK Σ_BL - η_BL Σ_AK + η_AL Σ_BK	True
107	the complex structure J = Γ17[8] commutes with all Σ17[A,B] (Spin(1,7) acts complex-linearly on ℝ^16 = ℂ^8)	True

5.2 A boost along x1 and a rotation in the x4–x5 plane

#	check	result
108	S(β) and R(θ) are the exponentials: dS/dβ == Σ17[0,1].S, S(0) == ID16, dR/dθ == Σ17[4,5].R, R(0) == ID16, and S.Sinv == R.Rinv == ID16	True
109	S(β).Γ17[K].S(β)^-1 == Σ_B Λboost[[B+1,K+1]] Γ17[B]; Λboost is the boost along x1	True
110	R(θ).Γ17[K].R(θ)^-1 == Σ_B Λrot[[B+1,K+1]] Γ17[B]; Λrot is the rotation in the x4-x5 plane	True
111	both Λ preserve the 7+1 metric: Transpose[Λ].η17.Λ == η17	True

5.3 The 7+1 Dirac operator squares to the 7+1 wave operator

#	check	result
112	(Γ17^A ∂_A)^2 Ψ == (∂0^2 - ∂1^2 - ... - ∂7^2) Ψ for 16 arbitrary functions of x0..x7	True

5.4 Spinors on the mass shell: real amplitudes

#	check	result
113	p is on the mass shell (p.η17.p == m^2) and pslash.pslash == m^2 ID16	True
114	(pslash - m) w == 0 has 8 linearly independent REAL solutions w	True
115	Ψ = w Exp[-I p.x] solves (I Γ17^A ∂_A - m) Ψ == 0	True

5.5 Real spinor fields: massless with Γ17, massive with the real 32×32 matrices

#	check	result
116	the matrices anticommuting with every Γ17[A] are exactly the multiples of Γ17[8], and (c Γ17[8])^2 == -c^2 ID16: no real J with J.J == +ID16	True
117	null momentum k = (1,1,0,0,0,0,0,0): k.η17.k == 0, kslash.kslash == 0, and kslash.w == 0 has 8 REAL solutions	True
118	Ψ = w Cos[k.x] is a REAL solution of the real massless Dirac equation Γ17^A ∂_A Ψ == 0	True
119	(Γ71R32^A ∂_A)^2 Ψ == -(∂0^2 - ∂1^2 - ... - ∂7^2) Ψ for 32 arbitrary functions of x0..x7	True
120	p.η71.p == -m^2 and pslash32.pslash32 == -m^2 ID32	True
121	Ψ = w Cos[p.x] + (pslash32.w/m) Sin[p.x], with 32 arbitrary REAL constants w, solves the REAL massive equation (Γ71R32^A ∂_A - m) Ψ == 0	True
122	... and Ψ obeys the Klein-Gordon equation (∂0^2 - ∂1^2 - ... - ∂7^2 + m^2) Ψ == 0	True

5.6 All nine signatures p + q = 8, and the corrected table of [1B]

#	check	result
123	split analogue: Re(x x) of split octonions == X.η53.X, and Γ53 realizes Cl(5,3) with REAL 16×16 matrices	True
124	Γ08[A] = T16O[8].T16O[A] realizes Cl(0,8) with REAL 16×16 matrices	True
125	gens[p] satisfy the Clifford relation of signature (p, 8-p) for p = 0..8	True
126	classification: ℝ(16) for (8,0),(5,3),(4,4),(1,7),(0,8); ℍ(8) for (7,1),(6,2),(3,5),(2,6)	True
127	every ℝ(16) case has explicit REAL 16×16 generators in this notebook (T16O, Γ53, T16, Γ17, Γ08)	True
128	ℂ(16) cannot occur: its real dimension 2*16^2 = 512 differs from 2^8 = 256	True

7.1 The eight real Dirac matrices and the Dirac α, β

#	check	result
129	Dirac matrices: Γ17up[0] == Γ17[0], Γ17up[i] == -Γ17[i]; all REAL, every entry 0 or ±1	True
130	Dirac adjoint: ConjugateTranspose[Γ17up[A]] == Γ17[0].Γ17up[A].Γ17[0] for A = 0..7	True
131	βD == {{0,ID8},{ID8,0}} and αD[i] == {{0,τO[i]},{-τO[i],0}}: real and symmetric	True
132	αD[i].αD[j] + αD[j].αD[i] == 2 δij ID16, αD[i].βD + βD.αD[i] == 0, βD.βD == ID16	True
133	ω17 = Γ17[8] == {{0,-ID8},{ID8,0}} is real, ω17.ω17 == -ID16, and ω17 anticommutes with every Γ17[A]	True

7.2 The field dirac16complex and its Dirac equation

#	check	result
134	(I Γ17up[A] ∂_A + m)(I Γ17up[A] ∂_A - m) dirac16complex == -(∂0^2 - ∂1^2 - ... - ∂7^2 + m^2) dirac16complex (16 arbitrary functions)	True

7.3 Hamiltonian form

#	check	result
135	Γ17[0].(I Γ17up[A] ∂_A - m) ψ == I ∂0 ψ - h ψ with h = -I αD[i] ∂_i + βD m (16 arbitrary functions)	True

7.4 Plane waves and the real spinors u_s, v_s

#	check	result
136	h(p) == {{0, M}, {Transpose[M], 0}} with the REAL 8×8 matrix M = m ID8 + p_i τO[i]	True
137	M.Transpose[M] == Transpose[M].M == E^2 ID8 and h(p).h(p) == E^2 ID16, E^2 = m^2 + p^2	True
138	h(p).u_s == E u_s and h(-p).v_s == -E v_s for s = 1..8	True
139	(pslash - m).u_s == 0 and (pslash + m).v_s == 0, pslash = E Γ17[0] + p_i Γ17[i]	True
140	u_s.u_r == v_s.v_r == 2E δsr, and u_s(p).v_r(-p) == 0	True
141	plane waves: u_s Exp[-I(E x0 - p.x)] and v_s Exp[+I(E x0 - p.x)] solve (I Γ17up[A] ∂_A - m) ψ == 0	True
142	example p = (2,2,4,0,0,0,0), m = 1: E = 5 and the eight u_s are real (entries in Sqrt[5] ℚ) and solve h(p).u_s == 5 u_s	True

7.5 Spin sums, completeness, and the diagonalization of h

#	check	result
143	spin sums: Σ_s u_s ubar_s == pslash + m and Σ_s v_s vbar_s == pslash - m	True
144	completeness: Σ_s u_s(p) u_s(p)^T + Σ_s v_s(-p) v_s(-p)^T == 2E ID16	True
145	U = (u_1..u_8, v_1(-p)..v_8(-p))/Sqrt[2E] is orthogonal and Transpose[U].h(p).U == diag(E ID8, -E ID8)	True

7.6 Grassmann algebra: anticommuting components, Lagrangian, conserved current

#	check	result
146	Grassmann: θ1 θ2 == -θ2 θ1, θ1 θ1 == 0, associativity, and ∂/∂θ2 (θ1 θ2 θ3) == -θ1 θ3	True
147	the three plane waves U1, U2, U3 solve the Dirac equation with m = 1	True
148	Grassmann field: ψ_a ψ_b + ψ_b ψ_a == 0 for all a, b (in particular ψ_a^2 == 0)	True
149	current conservation ∂_A (ψbar Γ17up[A] ψ) == 0 for the Grassmann-valued solution ψ = θ1 U1 + θ2 U2 + θ3 U3	True

7.7 Quantization: mode expansion and canonical anticommutators

#	check	result
150	canonical anticommutator from the mode algebra: {ψ_a(x0,x), ψ_b(x0,y)^†} == δab (1/V) Σ_p Exp[I p.(x - y)] (five momenta, symbolic x, y)	True

7.8 Charge conjugation

#	check	result
151	ψ -> ψc = ω17.ψ^*: (I Γ17up[A] ∂_A - m)(ω17.ψ^*) == ω17.((I Γ17up[A] ∂_A - m) ψ)^* for arbitrary ψ = ψre + I ψim	True
152	(pslash + m).ω17.u_s == 0 and ω17.u_s == Σ_r v_r (-Transpose[M]/E)_rs; ω17.ω17 == -ID16 (no Majorana spinors)	True

7.9 Lorentz covariance

#	check	result
153	boost at β = 2 log 2: S == 5/4 ID16 + 3/4 Γ17[0].Γ17[1], p' = Λ.p == (55/8, -41/8, 2, 4, 0, 0, 0, 0) is on shell, (pslash' - m).S.u_s(p) == 0	True

7.10 Chirality: no real chirality matrix, complex chiral projections

#	check	result
154	χ17 = I ω17: Hermitian, χ17.χ17 == ID16, anticommutes with every Γ17[A], commutes with every Σ17[A,B]	True
155	PL17, PR17 are complex (not real); Γ17[A].PL17 == PR17.Γ17[A]; ConjugateTranspose[PL17].Γ17[0] == Γ17[0].PR17	True
156	EL, ER: χ17.EL == -EL, χ17.ER == ER, EL^†.EL == ER^†.ER == ID8, EL^†.ER == 0, EL.EL^† == PL17, ER.ER^† == PR17, W unitary	True

#	check	result
157	φL == (ψup + I ψdown)/Sqrt[2], φR == (I ψup + ψdown)/Sqrt[2], PL17.ψ == EL.φL, PR17.ψ == ER.φR, ψ == EL.φL + ER.φR	True

7.11 The two coupled 8-component equations

#	check	result
158	σW[i] = I τO[i]: Hermitian, σW[i].σW[j] + σW[j].σW[i] == 2 δij ID8	True
159	chiral (Weyl) form: W^†.h(p).W == {{-σW.p, m ID8}, {m ID8, σW.p}} (symbolic p)	True
160	EL^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 + τO[i] ∂_i) φL - m φR (16 arbitrary functions)	True
161	ER^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 - τO[i] ∂_i) φR - m φL (16 arbitrary functions)	True
162	Klein-Gordon for each 8-component field: (I ∂0 - τO[i] ∂_i)(I ∂0 + τO[j] ∂_j) f == -(∂0^2 - ∂1^2 - ... - ∂7^2) f (8 arbitrary functions)	True
163	plane-wave example p = (2,2,4,0,0,0,0), m = 1, E = 5: (E + σW.p) φL == m φR and (E - σW.p) φR == m φL for all eight u_s	True

8.1 The self-coupled field: Lagrangian, densities, Hamiltonian

#	check	result
164	the densities ρS, ρP, j^A, j5^A are real: Γ17[0].op is Hermitian for op = ID16, I χ17, Γ17up[A], Γ17up[A].χ17	True
165	18 vertex pairs; lower vertex == η17[[A+1,A+1]] upper vertex in the V and A channels	True

8.2 The nonlinear field equation (Grassmann left derivatives)

#	check	result
166	nonlinear field equation: left derivative ∂L_int/∂ψbar_a == -[gS (ψbar ψ) ψ + gP (ψbar Iχ17 ψ) Iχ17 ψ + gV (ψbar Γ17[A] ψ) Γ17up[A] ψ + gA (ψbar Γ17[A]χ17 ψ) Γ17up[A]χ17 ψ]_a (32 Grassmann generators)	True

8.4 The 16-component Kohn–Sham equations

#	check	result
167	h_KS(p) is Hermitian for real S, P, V_A, A_A (symbolic momentum and potentials)	True
168	plane wave: Γ17[0].(pslash(ε,p) - m - S - I χ17 P - Γ17up[A](V_A + χ17 A_A)) == ε ID16 - h_KS(p)	True
169	Hartree potentials: ∂E_H/∂φ_a^* == (v_H.φ)_a, v_H = h_KS - h with S = gS ρS, P = gP ρP, V_A = gV j_A, A_A = gA j5_A (32 variables)	True

8.5 Exchange: Wick's theorem for contact couplings

#	check	result
170	Wick's theorem in the 560-dimensional Fock space: <Φ| :(ψbar o1 ψ)(ψbar o2 ψ): |Φ> == tr(o1 R) tr(o2 R) - tr(o1 R o2 R) for all 18 vertex pairs	True

8.6 The exchange self-energy and the Fierz identities

#	check	result
171	exchange self-energy: d/dt E_x(R + t dR) at t = 0 == tr(Σ_x.dR) and E_x == tr(Σ_x.R)/2 (random complex R, dR; symbolic couplings)	True
172	the 256 products Γ_K are orthogonal matrices with tr(Inverse[Γ_K].Γ_L) == 16 δKL	True
173	Fierz completeness: Σ_K (Γ_K)_ab (Inverse[Γ_K])_cd == 16 δad δcb (as the 256×256 matrix identity Transpose[F].F == 16 ID)	True
174	Fierz rearrangement of every exchange term: tr(o1 R o2 R) == (1/16) Σ_K tr(Inverse[Γ_K].R) tr(o1.Γ_K.o2.R) for all 18 vertex pairs (random complex R)	True

8.7 Example: uniform matter (no sea), Hartree and Hartree–Fock

#	check	result
175	uniform matter: tr R == ρS, tr(Γ17up[0].R) == ρ, and the densities ρP, j^i, j5^A vanish	True
176	the occupied plane waves give this R: (1/2) Σ_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) == (E Γ17[0] + m ID16)/(2E)	True
177	uniform exchange energy: E_x(R) == -gS/32 (ρS^2 + ρ^2) + gP/32 (ρS^2 - ρ^2) - gV/32 (8ρS^2 - 6ρ^2) + gA/32 (8ρS^2 + 6ρ^2)	True
178	uniform exchange self-energy: Σ_x == S_x ID16 + V_x Γ17[0] with S_x == ∂E_x/∂ρS and V_x == ∂E_x/∂ρ	True
179	uniform Hartree-Fock == Hartree with the Fierz-transformed couplings: gS/2 ρS^2 + gV/2 ρ^2 + E_x == G_S/2 ρS^2 + G_V/2 ρ^2 (gP, gA arbitrary)	True
180	c7 == 8 |S^6|/(2 Pi)^7 == 1/(15 Pi^4), |S^6| == 2 Pi^(7/2)/Gamma[7/2] == 16 Pi^3/15, and ρ == c7 kF^7/7 == kF^7/(105 Pi^4)	True
181	closed forms: d/dkF of the integrals gives the integrands, both vanish at kF = 0, and dε_kin/dm* == ρS	True
182	Hartree gap equation m* == m + gS ρS(m*) solved to 45 digits; m* is a stationary point of ε(m*) at fixed kF	True
183	total energy == sum of KS eigenvalues - E_H (double counting) == ε_kin - (m* - m) ρS + E_H	True
184	Hugenholtz-Van Hove at saturation: E/A == ε_F == dε/dρ; E/A - m < 0 and E/A at 0.95 kF0 and 1.05 kF0 is higher	True
185	Hugenholtz-Van Hove away from saturation: μ == dε/dρ == ε_F at kF = 1	True
186	Hartree-Fock at kF = 3/4 (gP = gA = 0): G_S == -98750, G_V == 20000; m* - m == gS ρS + S_x, V_0 == gV ρ + V_x; μ == ε_F	True

9.1 Chiral densities

#	check	result
187	chiral densities: ρS == κ + κ^*, ρP == I(κ - κ^*), j^A == jL^A + jR^A, j5^A == jR^A - jL^A (symbolic chiral orbitals)	True
188	the chiral blocks behind these densities: EL^†.βD.ER == ID8, EL^†.βD.EL == 0, EL^†.αD[i].EL == -σW[i], ER^†.αD[i].ER == σW[i], EL^†.αD[i].ER == 0	True

9.2 The Hartree energy in chiral variables

#	check	result
189	E_H (16-component form, chiral orbitals) == gS/2 (κ + κ^*)^2 - gP/2 (κ - κ^*)^2 + (gV + gA)/2 (jL.jL + jR.jR) + (gV - gA) jL.jR	True

9.3 The chiral Kohn–Sham equations

#	check	result
190	chiral KS Hamiltonian: W^†.h_KS(p).W == {{V^L_0 - σW.(p + V^L), M}, {M^*, V^R_0 + σW.(p + V^R)}}, M = m + S + I P, V^L = V - A, V^R = V + A	True
191	Hartree (Wirtinger): M - m == ∂e_H/∂κ == gS ρS + I gP ρP, V^L_A == ∂e_H/∂jL^A; V^L_A == V_A - A_A and V^R_A == V_A + A_A with V_A = gV j_A, A_A = gA j5_A	True
192	chiral KS potentials are derivatives: ∂E_H/∂φL^* == (V^L_0 - σW[i] V^L_i) φL + (M - m) φR and ∂E_H/∂φR^* == (V^R_0 + σW[i] V^R_i) φR + (M - m)^* φL	True

9.4 Chiral symmetry

#	check	result
193	e^(Iθχ17) == cosθ ID16 - sinθ ω17 is REAL and orthogonal; U.EL == e^(-Iθ) EL, U.ER == e^(Iθ) ER	True
194	U^†.βD.U == βD.(cos 2θ ID16 + I sin 2θ χ17); vector and axial densities invariant	True
195	E_H(e^(Iθχ17) ψ) - E_H(ψ) == (gS - gP)/2 [(e^(4Iθ) - 1) κ^2 + (e^(-4Iθ) - 1) κ^*2]: U(1)_A symmetric iff gP == gS	True

9.5 Uniform matter in chiral form

#	check	result
196	uniform matter in chiral form: κ == ρS/2, jL^0 == jR^0 == ρ/2, jL^i == jR^i == 0	True
197	(h_χ - V_0)^2 == (p^2 + m*^2) ID16: eigenvalues V_0 ± Sqrt[p^2 + m*^2], eight times each	True

9.6 The chiral vacuum with the Dirac sea: spontaneous chiral symmetry breaking

#	check	result
198	|S^6|/(2 Pi)^7 == 1/(120 Pi^4); J(0) == Λ^6/(720 Pi^4); g_c == -1/(8 J(0)) == -90 Pi^4/Λ^6; J(M) -> J(0) and K(M) -> K(0) as M -> 0	True
199	m = 0, g = 6/5 g_c: the gap equation 1 + 8 g J(|M|) == 0 has the root |M| (32 digits); M == 2 g κ^* with κ == -4 M J; E is stationary there and lower than E(0)	True
200	degenerate chiral vacua for m = 0: E(|M| e^(I ϕ)) == E(|M|) for ϕ = 1, 2, 4 (the phase is the Goldstone direction)	True
201	explicit breaking m = 1/100: the real M == m - 8 g J(M) M solves the gap equation, E is stationary there, and every complex phase raises E	True
202	the sea at one momentum: h_χ^2 == E^2 ID16; Π_- is a projector of rank 8; κ_p == tr(Π_- RL block) == -4 M^*/E; tr(Π_-.h0) == -8E + 8 Re[(M - m) M^*]/E	True

4.3 Independent Python cross-checks

tools/verify_7p1_nb.py (parses the evaluated 7+1.nb; exit code 0)

```
input cells
    ...: 92
output cells
    ...: 241
message cells (must be 0)
    ...: 0
check outputs
```

```
    ...: 202
check outputs that are True
    ...: 202
check["..."] calls in the input cells
    ...: 202
$Version output
    ...: ['{"15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)","Notebook$$38$50910`"}']
private notebook context
    ...: True
Length[checks]
    ...: 202
failedChecks == {}
    ...: True
allChecksPassed output True
    ...: True
tau[A] printed == 4+4 tau[A] (python)
    ...: True
tauO[A] printed == python tauO[A]
    ...: True
Gamma17[A] printed == python G17[A]
    ...: True
gridO printed == python octonion table
    ...: True
gridx44 printed == python split table (4+4.nb)
    ...: True
Lambda boost/rotation printed
    ...: True
8 real on-shell spinors printed solve (pslash - m) w = 0 (python check)
    ...: True
classification table printed
    ...: True
section 7: check outputs (all True)
    ...: 35 (True)
section 8: check outputs (all True)
    ...: 23 (True)
section 9: check outputs (all True)
    ...: 16 (True)
section 7: printed Sqrt[5] u_s(p) == python (5 e_s; M^T e_s) and (pslash - m) u_s == 0
    ...: True
section 8: printed closed forms of int k^6 m*/E and int k^6 E: d/dkF == integrands, zero at kF
    ...= 0 (sympy) : True
sections 8-9: 23 printed numbers agree with tools/ref_dirac16.py to 17 digits
    ...: True
sections 8-9: G_S == -98750, G_V == 20000 printed
    ...: True
tools/ref_dirac16.py: all its boolean checks True
    ...: True
sections 0-6: the 166 input and text cells equal those of the backup
    ...: True

VERDICT: ALL CROSS-CHECKS PASSED
```

tools/ref44.py (rebuilds 4+4.nb; exit code 0)

```
ζ7 = [-1, -1, -1, -1, 1, 1, 1, 1]
σ == ζ1ζ2ζ3 : True
{ζA,ζB}/2 == -η ID8 (A,B=1..7): True
-η_AA ζA == ζA^T : True
(τA τB+τB τA)/2 == η ID8 : True
(τA τB+τB τA)/2 == η ID8 : True
τ1..τ7 product == ID8: True
σ == τ4τ5τ6τ7: True
{T16A,T16B}/2 == η ID16 : True
```

```
T16[8] diag: [-1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1]
σ16 == diag(-σ,σ): True
unit: [sqrt(2)/2, 0, 0, 0, sqrt(2)/2, 0, 0, 0]  unit.σ.unit = 1
E^a_A[h] == (u σ u) Transpose[Table[η_BB τ[B-1].u]] for all h: True
eA[1] : ['eA[1]', 'eA[2]', 'eA[3]', 'eA[4]', 'eA[5]', 'eA[6]', 'eA[7]', 'eA[8]']
eA[2] : ['eA[2]', '-eA[1]', '-eA[4]', 'eA[3]', '-eA[6]', 'eA[5]', 'eA[8]', '-eA[7]']
eA[3] : ['eA[3]', 'eA[4]', '-eA[1]', '-eA[2]', '-eA[7]', '-eA[8]', 'eA[5]', 'eA[6]']
eA[4] : ['eA[4]', '-eA[3]', 'eA[2]', '-eA[1]', '-eA[8]', 'eA[7]', '-eA[6]', 'eA[5]']
eA[5] : ['eA[5]', 'eA[6]', 'eA[7]', 'eA[8]', 'eA[1]', 'eA[2]', 'eA[3]', 'eA[4]']
eA[6] : ['eA[6]', '-eA[5]', 'eA[8]', '-eA[7]', '-eA[2]', 'eA[1]', '-eA[4]', 'eA[3]']
eA[7] : ['eA[7]', '-eA[8]', '-eA[5]', 'eA[6]', '-eA[3]', 'eA[4]', 'eA[1]', '-eA[2]']
eA[8] : ['eA[8]', 'eA[7]', '-eA[6]', '-eA[5]', '-eA[4]', '-eA[3]', 'eA[2]', 'eA[1]']
```

tools/ref71.py (rebuilds 7+1.nb objects; exit code 0)

```
tauO antisymmetric (A>=1)                          : True
tauO[A]^2=-1 (A>=1)                                : True
tauO Euclid relation                               : True
tauObar tauO relation                              : True
tauO1..7 product                                   : [1, 1, 1, 1, 1, 1, 1, 1]
sigma commutes tau1..3, anticommutes tau4..7       : True
E orthogonal                                       : True
oct: unit two-sided                                : True
oct: e_i^2                                         : [-1, -1, -1, -1, -1, -1, -1]
oct: x xbar = N(x) 1                               : True
oct: N(xy)=N(x)N(y)                                : True
oct: conj(xy)=conj(y)conj(x)                       : True
oct: alternative (left,right,flexible)             : True
oct: Moufang identities                            : True
oct: #nonassociative basis triples                 : 168
oct: #noncommuting ordered basis pairs             : 42
e_A*e_B (O) = s_AB e_A*e_B (S)                     : True
T16O Euclid Cl(8,0)                                : True
T16O[A]==T16A44[A] (A<=3)                          : True
T16O[k]==T16A44[k].S44 (k>=4)                      : True
S44==diag(sigma,sigma)                             : True
T16O[8] diag                                       : [-1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1,
    ...1, 1, 1, 1, 1]
G17 Cl(1,7)                                        : True
G17 real                                           : True
G17[0] sym, G17[i] antisym                         : True
G17[8]^2                                           : -1
G17 blocks: G17[i]==diag(tauO,tauObar)             : True
G71 Cl(7,1) eta71                                  : True
G71[8]^2                                           : -1
G71 = Ssim (I G17) Ssim^-1                         : True
G71R32 Cl(7,1) real                                : True
```

tools/extra71.py (further identities; exit code 0)

```
Lambda boost (rows B = 0..7):
    ['cosh(beta)', '-sinh(beta)', '0', '0', '0', '0', '0', '0']
    ['-sinh(beta)', 'cosh(beta)', '0', '0', '0', '0', '0', '0']
    ['0', '0', '1', '0', '0', '0', '0', '0']
    ['0', '0', '0', '1', '0', '0', '0', '0']
    ['0', '0', '0', '0', '1', '0', '0', '0']
    ['0', '0', '0', '0', '0', '1', '0', '0']
    ['0', '0', '0', '0', '0', '0', '1', '0']
    ['0', '0', '0', '0', '0', '0', '0', '1']
Lambda rotation (rows B = 0..7):
    ['1', '0', '0', '0', '0', '0', '0', '0']
```

```
    ['0', '1', '0', '0', '0', '0', '0', '0']
    ['0', '0', '1', '0', '0', '0', '0', '0']
    ['0', '0', '0', '1', '0', '0', '0', '0']
    ['0', '0', '0', '0', 'cos(theta)', '-sin(theta)', '0', '0']
    ['0', '0', '0', '0', 'sin(theta)', 'cos(theta)', '0', '0']
    ['0', '0', '0', '0', '0', '0', '1', '0']
    ['0', '0', '0', '0', '0', '0', '0', '1']
indicator classification (signature, number of intertwiners, M.conj(M) for the normalized M,
    ...algebra):
    Cl(8,0): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(7,1): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(6,2): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(5,3): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(4,4): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(3,5): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(2,6): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(1,7): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(0,8): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
EO == eta4488 . E44[8]                                              : True
G53 Cl(5,3) real                                                    : True
G08 Cl(0,8) real                                                    : True
so(1,7): [S_AB,S_CD]=eta_BC S_AD - eta_AC S_BD - eta_BD S_AC + eta_AD S_BC: True
[S_AB,G_C]=eta_BC G_A - eta_AC G_B                                  : True
S Sinv = 1                                                          : True
S G_C S^-1 == sum_B L[B,C] G_B                                      : True
L^T eta L == eta                                                    : True
p on shell p.eta17.p == 1                                           : True
pslash^2 == 1                                                       : True
8 real solutions                                                    : True
classification                                                      : True
explicit quaternion commutant of G71R32                             : True
G17e[0]==[[0,1],[1,0]]x1, G17e[i]==diag(LO[i],-LO[i])               : True
Gamma17(x)^2 == (x.eta17.x) 1                                       : True
anticommutant of G17: dimension 1 (numerical rank 255 of 2048x256 system): True
G17[8] anticommutes with all G17[A], G17[8]^2 == -1                 : True
null k: k.eta17.k == 0, kslash^2 == 0, 8 rational solutions of kslash w = 0: True
Psi = w cos(k.x) is a REAL solution of G17^A d_A Psi = 0            : True
p.eta71.p == -m^2 and P32^2 == -m^2                                 : True
Psi32 (32 arbitrary real w) solves (G71R32^A d_A - m) Psi = 0       : True
Psi32 obeys the Klein-Gordon equation (box17 + m^2) Psi = 0         : True
(G71R32^A d_A)^2 == -box17 on 32 arbitrary functions                : True
```

tools/ref_dirac16.py (independent checks of sections 7-9; exit code 0)

```
Gamma17 real, entries 0,+-1 : True
{Gamma_A,Gamma_B} = 2 eta17_AB : True
Gamma0 symmetric, Gamma_i antisymmetric : True
Gamma^A dagger = Gamma0 Gamma^A Gamma0 : True
alpha^i, beta real symmetric : True
{alpha_i,alpha_j}=2delta, {alpha_i,beta}=0, beta^2=1 : True
block forms alpha^i = [[0,tauO_i],[-tauO_i,0]], beta = [[0,1],[1,0]] : True
omega = Gamma17[8] = [[0,-1],[1,0]], omega^2 = -1 : True
(i Dslash + m)(i Dslash - m) psi = -(box + m^2) psi : True
Gamma0 (i d0 - h) psi = (i Dslash - m) psi : True
h(p) = [[0,M],[M^T,0]] : True
h(p)^2 = E^2 1 : True
M M^T = M^T M = E^2 1_8 : True
h(p) u_s = E u_s,  h(-p) v_s = -E v_s : True
(pslash - m) u_s = 0, (pslash + m) v_s = 0 : True
u_s^T u_r = 2E delta, v_s^T v_r = 2E delta : True
u_s(p)^T v_r(p~) = 0 : True
sum_s u_s ubar_s = pslash + m, sum_s v_s vbar_s = pslash - m : True
completeness sum u u^T + sum v(p~) v(p~)^T = 2E 1_16 : True
```

```
U_p orthogonal and U_p^T h(p) U_p = diag(E 1_8, -E 1_8) : True
charge conjugation: (pslash + m) omega u_s = 0 : True
omega u_s = sum_r v_r (-M^T/E)_rs : True
S Gamma_K S^-1 = sum_B Lambda_BK Gamma_B (boost) : True
Lorentz covariance: (pslash(Lambda p) - m) S u_s(p) = 0 : True
real matrices anticommuting with all Gamma17[A]: dimension 1 (rank 255) : True
omega anticommutes with all Gamma17[A] : True
chi = i omega: Hermitian, chi^2 = 1, {chi,Gamma_A} = 0 : True
[chi, Sigma_AB] = 0 (chirality is Lorentz invariant) : True
P_L, P_R not real : True
P_L+P_R = 1, P_L^2 = P_L, P_R^2 = P_R, P_L P_R = 0, rank 8 each : True
Gamma_A P_L = P_R Gamma_A,  P_L^dagger Gamma0 = Gamma0 P_R : True
E_L, E_R isometries: E^dag E = 1_8, E_L^dag E_R = 0, E_L E_L^dag = P_L, E_R E_R^dag = P_R : True
chi E_L = -E_L, chi E_R = E_R, W unitary : True
W^dag h(p) W = [[-sigma.p, m],[m, sigma.p]] : True
sigma_i Hermitian, {sigma_i,sigma_j} = 2 delta : True
chiral blocks: E_L^dag beta E_L = 0, E_L^dag beta E_R = 1, E_L^dag alpha E_L = -sigma, E_R^dag
    ...alpha E_R = sigma, E_L^dag alpha E_R = 0 : True
E_L^dag Gamma0 (i Dslash - m) psi = (i d0 + tauO.d) phi_L - m phi_R : True
E_R^dag Gamma0 (i Dslash - m) psi = (i d0 - tauO.d) phi_R - m phi_L : True
(i d0 - tauO.d)(i d0 + tauO.d) f = -box f (8 components) : True
psi = E_L phi_L + E_R phi_R : True
chiral form of the bilinears (beta i chi, chi, alpha chi) : True
{psi_a(t,x), psi_b^dag(t,y)} = delta_ab (1/V) sum_q e^{iq(x-y)} (5 momenta, abstract mode
    ...algebra) : True
{psi_a(t,x), psi_b(t,y)} = 0 : True
the three U_k solve the Dirac equation (m = 1) : True
Grassmann components anticommute: psi_a psi_b + psi_b psi_a = 0, psi_a^2 = 0 : True
current conservation d_A (psibar Gamma^A psi) = 0 (Grassmann field) : True
dL_int/dpsibar_a = -[gS(psibar psi) psi + gP (psibar i chi psi) i chi psi + gV (psibar G_A psi)
    ...G^A psi + gA (psibar G_A chi psi) G^A chi psi]_a : True
h_KS Hermitian for real S, P, V_A, A_A : True
Gamma0 [pslash(eps,p) - m - S - i chi P - Gamma^A (V_A + chi A_A)] = eps - h_KS(p)  (plane wave
    ...e^{-i(eps t - p.x)}) : True
W^dag h_KS W = [[V^L_0 - sigma.(p+V^L), M],[M^*, V^R_0 + sigma.(p+V^R)]], V^L = V - A, V^R = V
    ...+ A, M = m + S + iP : True
Hartree potentials: dE_H/dphi*_a = (v_KS phi)_a with S = gS rhoS, P = gP rhoP, V_A = gV j_A,
    ...A_A = gA j5_A : True
densities real: Gamma0 O Hermitian for O = 1, i chi, Gamma^A, Gamma^A chi : True
chiral densities: rhoS = kappa + kappa^*, rhoP = i(kappa - kappa^*), j = jL + jR, j5 = jR - jL
    ...: True
E_H = gS/2 (k+k*)^2 - gP/2 (k-k*)^2 + (gV+gA)/2 (jL.jL + jR.jR) + (gV-gA) jL.jR : True
V^L_A = V_A - A_A, V^R_A = V_A + A_A (Hartree, chiral currents) : True
M - m = dE_H/dkappa = gS(k+k*) - gP(k-k*) = S + iP : True
dE_H/dphi_L* = (V^L_0 - sigma_i V^L_i) phi_L + (M - m) phi_R ; dE_H/dphi_R* = (V^R_0 + sigma_i
    ...V^R_i) phi_R + (M - m)^* phi_L : True
U_theta = cos(theta) - sin(theta) omega is real orthogonal; U E_L = e^{-i theta} E_L, U E_R =
    ...e^{i theta} E_R : True
U^dag beta U = beta (cos 2theta + i sin 2theta chi);  vector and axial currents invariant : True
E_H(e^{i theta chi} psi) - E_H(psi) = (gS-gP)/2 [(e^{4i theta}-1) k^2 + (e^{-4i theta}-1) k*^2]
    ...->  U(1)_A symmetric iff gP = gS : True
256 products Gamma_K are orthogonal: Gamma_K^-1 = Gamma_K^T : True
tr(Gamma_K^-1 Gamma_L) = 16 delta_KL (256 x 256) : True
Fierz completeness: sum_K (Gamma_K)_ab (Gamma_K^-1)_cd = 16 delta_ad delta_cb : True
delta E_x = tr[Sigma_x delta R] (random R, dR), Sigma_x = -gS R - gP (i chi) R (i chi) - gV G_A
    ...R G^A - gA G_A chi R G^A chi : True
E_x = (1/2) tr[Sigma_x R] : True
E_H = (gS/2)(tr R)^2 + ... = sum_c (g_c/2) tr(O_c R) tr(O^c R) and dE_H = tr[Sigma_H dR] : True
Fierz: tr[O1 R O2 R] = (1/16) sum_K tr(Gamma_K^-1 R) tr(O1 Gamma_K O2 R) for all channel
    ...vertices : True
uniform R: tr R = rho_S, tr(Gamma^0 R) = rho : True
(1/2) sum_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) = (E Gamma_0 + m)/(2E) : True
E_x(uniform) = -gS/32 (rS^2+rho^2) + gP/32 (rS^2-rho^2) - gV/32 (8rS^2-6rho^2) + gA/32
    ...(8rS^2+6rho^2) : True
Sigma_x(uniform) = S_x + V_x Gamma_0, S_x = rS/16(-gS+gP-8gV+8gA), V_x = rho/16(-gS-gP+6gV+6gA)
```

```
    ...: True
S_x = dE_x/drho_S, V_x = dE_x/drho : True
uniform Hartree-Fock = Hartree with G_S = gS + (-gS+gP-8gV+8gA)/16, G_V = gV +
    ...(-gS-gP+6gV+6gA)/16 : True
uniform matter: Hartree terms of P and A vanish (rho_P = 0, j5 = 0, j^i = 0) : True
Wick (3 particles, 16 modes): <:(psibar O psi)(psibar O' psi):> = tr(O R) tr(O' R) - tr(O R O'
    ...R) for all 18 vertex pairs : True
c7 = 8 |S^6|/(2pi)^7 = 1/(15 pi^4) and rho = c7 kF^7/7 = kF^7/(105 pi^4) : True
NUM Hartree saturation kF0 : 0.75663615070149289469
NUM Hartree rho0 : 0.00001388100213231687719 7
NUM Hartree m* : 0.34809734014624326514
NUM Hartree rhoS : 6.5190265985375673486e-6
NUM Hartree V0 : 0.13881002132316877197
NUM Hartree E/A - m : -0.028321427964738575854
NUM Hartree eps_F : 0.97167857203526142415
Hartree gap: m* - m - gS rhoS(m*) = 0 : True
m* is a stationary point of eps(m*) = ekin(m*) - (m*-m)^2/(2gS) + gV rho^2/2 : True
total energy = sum eps_k - E_H (double counting) = ekin - (m*-m) rhoS + E_H : True
Hugenholtz-Van Hove at saturation: E/A = eps_F = d eps/d rho : True
d(eps/rho)/dkF = 0 at kF0 (minimum: E/A(0.95 kF0), E/A(1.05 kF0) larger) : True
Hugenholtz-Van Hove mu = d eps/d rho = eps_F at kF = 1 (not saturated) : True
NUM Hartree kF=1: m* : 0.080964430277606126329
NUM Hartree kF=1: E/A - m : 0.41086484863433217115
NUM Hartree kF=1: eps_F : 1.9809848613081542876
NUM HF couplings G_S, G_V : -98750.0, 20000.0
HF self-consistency with the exchange self-energy: m* - m = gS rhoS + S_x, V0 = gV rho + V_x :
    ...True
HF energy: ekin - (m*-m) rhoS + E_H + E_x = eps(G_S, G_V) : True
HF Hugenholtz-Van Hove: d eps/d rho = eps_F at kF = 3/4 : True
NUM HF kF=3/4: m* : 0.36643630198154392124
NUM HF kF=3/4: V0 : 0.2610177547316155257
NUM HF kF=3/4: E/A - m : 0.039146338016119861526
NUM Hartree kF=3/4: m* : 0.36297394764803019822
NUM Hartree kF=3/4: E/A - m : -0.028065375846872960879
J(0) = Lambda^6/(720 pi^4), g_c = -1/(8 J(0)) = -90 pi^4/Lambda^6 : True
NUM NJL g_c : -8766.8181930602193513
NUM NJL g = 6/5 g_c : -10520.181831672263222
NUM NJL |M| (m = 0) : 0.55369385182924169122
NUM NJL kappa = -4 M J(|M|) (M real) : -0.00002631579285836487 42
NUM NJL E(|M|) - E(0) : -1.3102449629010022473e-6
NJL m = 0: gap 1 + 8 g J(|M|) = 0 solved, M = 2 g kappa^*, stationary and lower than M = 0 :
    ...True
NJL m = 0: E(|M| e^{i theta}) independent of theta (degenerate chiral vacua) : True
NUM NJL M (m = 1/100) : 0.58531284674622051123
NJL m = 1/100: M = m - 8 g J(M) M real, stationary; E(M e^{i theta}) > E(M) for theta != 0 :
    ...True
sea at one momentum: eigenvalues -E (x8), +E (x8); kappa_p = tr[(Pi_-)_{RL}] = -4 M^*/E;
    ...tr[Pi_- h_0] = -8E + 8 Re[(M-m) M^*]/E : True
boolean checks: 90   failed: 0
VERDICT: ALL CHECKS PASSED
```

4.4 Hashes of the files used to build 7+1.nb and its documentation

file	SHA-256
7+1.nb	438d972073e2fef9d7da6d57a541432b0fb1ea08424307f59ae2f3f7752bbfaf
4+4.nb	740023233eaa0cd24ca0e2e7010094671b1d314b698114c08db94050d3e0aed5
tools/cayley_dickson_formulas.py	bdde8c4c8199b09ef36ae4e15d5d53bc62744fc85b4c5ba52eab5516f77b5d5f
tools/cayley_dickson_match.py	73f2cc584bbe1a3620c50890ffe5fb299857b04a14bc34fed2f9eaefd2e36c4e
tools/classify_clifford.py	42291276f46b046183fdbb7b62be7c5cedfd695fe9da96ac256afa23580930a9
tools/doc_common.py	cc873e042a996ea0f1078239656771fb450b5f8cf81ed53d63f0373e7d35e3ba
tools/doc_dirac16.py	eceeb4921a73f08a21ca01ad89e71ee9a98e89704d033725a39caf1f870cc850

file	SHA-256
tools/doc_dirac16_common.py	9324b4fae59e51019125ebe09157f3c8c9489560e1383a697fe6c22a3a93e208
tools/doc_dirac16_provenance.py	62a270d53007c3d54e82433ea537c053ba6d250181d03e763a73f38a690d933f
tools/doc_dirac16_symbols.py	70cee4b84501c0f7a7a02105e39acfa6e21c61835a667eca28e43e6ad65cf674
tools/doc_provenance.py	027090f677e384b3c714f68f6471ebf9bd1d3babccb8a168c03a324ad6e72eea
tools/doc_readme.py	972c16fb66070aa0e9c21b3dc2ec3403182a505ac1537065fce48854849d0cf4
tools/doc_symbols.py	def89bf1b3935a7e95651f6a76bddb61242c840414676da4c8d7a393e6da1541
tools/docgen.py	bd229bca182d2de5939c5af0e9879e6ee22721629ee45518490b118661d76cfd
tools/eval_cells_log.wl	833b6bd854cab253e5bb935c28f28ab50b286f409d130f433fa5dfb9b9236bbe
tools/extra71.py	9935ef99cd9300f2a7d4f06d36cf7e173912ebae61b6feff448ebe0588853168
tools/git_push.wl	ac05a6d0256834b48cc9fa943fa247bccc574f723e68f053d00bfba9aca2e21a
tools/make_7p1_nb.py	09214fb4b0d8901cf6b2edde09b3f048d899721ed4e269bad98d334ba9b6d1af
tools/make_docs.py	7231ed786fa15f5d7b81963971fa88ef35bb08e0166f0af4313c5a23e64bcbd5
tools/nbparse.py	ad2255a8864be7447a55023840b327f789cb8c9d868d42261b19261653847423
tools/nbwrite.py	dcbe8ce5b3203c89a1d17fbe946e1b6b7bb7f8895afa944743d651cfefadeffb
tools/octalg.py	36b50120b69a963e252e07d60ec007755220822b4ad339716ea7aed93d58eb24
tools/ref44.py	ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064faf5
tools/ref71.py	4ac783b1ab221317387f2a74e0be60f114fd64940fe3ea96af79b33ee7dcb8b0
tools/ref_dirac16.py	a6e139ebe5199621a95c8adc772bc3e1e9b452c0ca20a963c536cbfc6539319c
tools/run_eval.wl	2c87cf88662c3932e17de1e7b67c9910e2b14da01e4e78f4529873cef93c89eb
tools/verify_7p1_nb.py	52f54cf0e110f4a22ac4612619cc74f32ab98c68fd372b786fb6ba943dfe2dbf
tools/watch_requests.wl	d9ceedb0b45ac3cf9b4dd8519ac164953f2c80e5188b1557fe20ef5107ae352d

5. Process and environment

- Workspace: `D:\Developer\github\7+1` on Windows; Python and LaTeX ran in a Linux shell with that folder mounted. Intermediate files go to `build/`, which the repository's `.gitignore` excludes.
- The notebook is generated by `tools/make_7p1_nb.py` (cells written with `tools/nbwrite.py`), evaluated in place by `tools/run_eval.wl` in Wolfram 15.0.1, and cross-checked by `tools/verify_7p1_nb.py`.
- Python 3.10.12 with sympy 1.14.0 and numpy 2.2.6 (exact rational arithmetic for every identity).
- The documentation is generated by `tools/make_docs.py` from the modules `docgen.py`, `doc_common.py`, `doc_dirac16_common.py`, `doc_readme.py`, `doc_symbols.py` and `doc_provenance.py` of the folder tools/ (and the number formatting of `doc_dirac16.py`): one source renders Markdown and LaTeX, and every matrix, table and check list is computed or parsed, never typed. The LaTeX is compiled with XeLaTeX (TeX Live 2022, `latexmk -xelatex`, fonts Latin Modern and DejaVu Sans Mono).
- History: the first algebraic experiments were made in a temporary cloud sandbox and then moved into `tools/` of the workspace, where everything was re-run. A Wolfram 15.0 kernel window was opened once by mistake before Wolfram 15.0.1 was available; no result comes from it. Only Wolfram 15.0.1 evaluated 7+1.nb.
- Corrections made during the review: an earlier draft called the real on-shell amplitudes of notebook section 5.4 "Majorana spinors"; that label was removed, and section 5.5 was added to show which real spinor FIELDS exist (massless with 16 components, massive only with the 32×32 matrices).
- The evaluated notebook was rendered section by section in the Wolfram 15.0.1 front end (`Rasterize`) and inspected. Characters that the front end's text font shows badly are written differently in text cells: the double-struck O and the arrow of $\vec{x}$ as small inline formulas, the script ℓ in the font Cambria Math, and $\not{p}$, $\not{k}$ as `pslash`, `kslash`.
- Second stage (sections 7–10 of 7+1.nb): all Wolfram Language work ran in the front end `C:\Program Files\Wolfram Research\Wolfram\15.0.1\WolframNB.exe`. Jobs were requested without keyboard input through `tools/watch_requests.wl` (loaded once in that front end; every 10 seconds it reads `build/request.txt` and evaluates `tools/run_eval.wl`, `tools/git_push.wl` or a development file); new cells were developed with `tools/eval_cells_log.wl`, which evaluates a generated notebook expression by expression, logs messages and checks, and stops at the first failure. `tools/run_eval.wl` evaluates with a time limit of one hour. Pushes run the local Git for Windows from the front end (`tools/git_push.wl`, no login prompt) to `https://github.com/once-ere/7plus1.git`, the new name of the repository.

- Second stage, Python: `tools/ref_dirac16.py` (sympy, mpmath, numpy) is the independent reference; `tools/verify_7p1_nb.py` was extended to sections 7–9 and to the comparison of sections 0–6 with the backup of the first stage.

6. The second stage: sections 7–10 of 7+1.nb

This section records the provenance of the Dirac equation and the Kohn–Sham theories of the field dirac16complex (sections 7–10 of 7+1.nb). Notation: $\Gamma_A = \Gamma_A^{(1,7)}$, $\Gamma^A = \eta^{AA}_{(1,7)}\Gamma_A$, $\Gamma^0 = \Gamma_0$, $\chi = \chi^{(1,7)} = i\Gamma_8^{(1,7)}$, Ψ the field dirac16complex, $\bar\Psi = \Psi^\dagger\Gamma^0$, $\vec x = (x_1, \ldots, x_7)$, $\vec p = (p^1, \ldots, p^7)$.

6.1 The task

1. Back up 7+1.nb and append to it; the first part stays unchanged.
2. A checked Dirac equation on the real 7+1 spacetime for a 16-component quantized fermion field named dirac16complex whose components are complex-valued Grassmann operators, with eight PURELY REAL anticommuting 16×16 Dirac matrices taken from 7+1.nb.
3. Chiral projections writing the 16-component equation as two 8-component equations, coupled when the mass is not zero.
4. A 16-component Kohn–Sham theory of the field coupled to itself, and a chiral version using the chiral projections.
5. Complete, checked documentation (every document with a provenance file and a symbol file, Markdown and LaTeX, compiled to PDF, each self-contained); every completed stage pushed to github.com/once-ere/7plus1 (the repository was renamed from 7+1 to 7plus1); only the Wolfram front end 15.0.1 is used.

6.2 The first part of the notebook is unchanged

The evaluated notebook of the first stage was copied, in the workspace, to the file below (the folder backups/ is excluded from the repository by its .gitignore). `tools/verify_7p1_nb.py` compares every input, text and section cell of sections 0–6 of the new 7+1.nb with that copy: all are identical.

```
backups/7+1_2026-10-26_before_dirac16complex.nb
306963 bytes
SHA-256 f7abe7279b5540e914de876dc055c199bd491b9c2d9522bf28dbc23689e237ce
```

6.3 Derivations and decisions

Dirac matrices. The real generators $\Gamma_A^{(1,7)}$ of section 2.5 are used unchanged: $\Gamma_0 = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, $\Gamma_i = \left(\begin{smallmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{smallmatrix}\right)$, $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}^{(1,7)}1_{16}$. The opposite sign convention $\eta^{(7,1)}$ has no real 16×16 realization (section 2.7), so this is the only real choice. **The factor i.** With real Γ_A, $(\Gamma^A\partial_A)^2 = \frac{1}{2}\{\Gamma^A,\Gamma^B\}\partial_A\partial_B = \partial_0^2 - \partial_1^2 - \cdots - \partial_7^2$, hence

$$\big(i\Gamma^A\partial_A + m\big)\big(i\Gamma^A\partial_A - m\big) = -\big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 + m^2\big)1_{16},$$
$$\Gamma^0\big(i\Gamma^A\partial_A - m\big) = i\partial_0 - H_{\rm D}, \qquad H_{\rm D} = -i\Gamma^0\Gamma^i\partial_i + m\Gamma^0.$$

Real amplitudes. For $e^{-i(Ex_0 - \vec p\cdot\vec x)}$, with $P_\tau = p^i\tau_i^O$ ($P_\tau^{\mathsf T} = -P_\tau$, $P_\tau^2 = -\vec p^{\,2}1_8$):

$$H_{\rm D}(\vec p) = \begin{pmatrix} 0 & m1_8 + P_\tau \\ m1_8 - P_\tau & 0 \end{pmatrix}, \qquad (m1_8 + P_\tau)(m1_8 - P_\tau) = (m^2 + \vec p^{\,2})1_8 = E^2 1_8,$$

$$H_{\mathrm{D}}(\vec{p})\begin{pmatrix} E\,e_s \\ (m1_8 - P_\tau)e_s\end{pmatrix} = \begin{pmatrix} E^2 e_s \\ E(m1_8 - P_\tau)e_s\end{pmatrix} = E\begin{pmatrix} E\,e_s \\ (m1_8 - P_\tau)e_s\end{pmatrix},$$

$$w_s = \frac{1}{\sqrt{E}}\begin{pmatrix} E\,e_s \\ (m1_8 - P_\tau)e_s\end{pmatrix}, \qquad w_s^{\mathsf{T}} w_r = 2E\,\delta_{sr}.$$

Chirality. The real matrices anticommuting with every Γ_A are $c\,\Gamma_8^{(1,7)}$ (section 2.9), with square $-c^2 1_{16}$; the chirality is therefore the complex $\chi = i\Gamma_8^{(1,7)} = \left(\begin{smallmatrix} 0 & -i1_8 \\ i1_8 & 0\end{smallmatrix}\right)$. With $E_\pm$ the 16×8 matrices $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1_8 \\ -i1_8\end{smallmatrix}\right)$ and $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} -i1_8 \\ 1_8\end{smallmatrix}\right)$ ($\chi E_- = -E_-$, $\chi E_+ = E_+$, $E_- E_-^\dagger = P_L^{(1,7)}$, $E_+ E_+^\dagger = P_R^{(1,7)}$), $\Psi_L = E_-^\dagger \Psi$, $\Psi_R = E_+^\dagger \Psi$ and

$$E_-^\dagger \Gamma^0 E_+ = E_+^\dagger \Gamma^0 E_- = 1_8, \qquad E_\mp^\dagger \Gamma^0 E_\mp = 0, \qquad E_\mp^\dagger \Gamma^0 \Gamma^i E_\mp = \mp i\tau_i^O, \qquad E_-^\dagger \Gamma^0 \Gamma^i E_+ = 0,$$

$$E_-^\dagger \Gamma^0 \big(i\Gamma^A \partial_A - m\big)\Psi = \big(i\partial_0 + \tau_i^O \partial_i\big)\Psi_L - m\Psi_R, \qquad E_+^\dagger \Gamma^0 \big(i\Gamma^A \partial_A - m\big)\Psi = \big(i\partial_0 - \tau_i^O \partial_i\big)\Psi_R - m\Psi_L.$$

Self-coupling. The four Lorentz-invariant contact interactions without derivatives used in relativistic point-coupling models (scalar, pseudoscalar, vector, axial vector) with real couplings, and real sources that define the densities:

$$\begin{aligned}\mathcal{L} = {} & \bar\Psi\big(i\Gamma^A\partial_A - m\big)\Psi - \bar\Psi\big(\mathsf{s} + i\chi\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\mathsf{a}_A)\big)\Psi \\ & - \frac{g_S}{2}(\bar\Psi\Psi)^2 - \frac{g_P}{2}(\bar\Psi i\chi\Psi)^2 - \frac{g_V}{2}(\bar\Psi\Gamma^A\Psi)(\bar\Psi\Gamma_A\Psi) - \frac{g_A}{2}(\bar\Psi\Gamma^A\chi\Psi)(\bar\Psi\Gamma_A\chi\Psi).\end{aligned}$$

The tensor and higher-rank channels are not included. With $\hbar = c = 1$ every g has mass dimension -6 (an effective theory: no-sea approximation, or a momentum cutoff for the sea). **Hohenberg–Kohn.** For $H[w] = H_0 + H_{\mathrm{int}} + \int w\cdot n$ with $w\cdot n = \mathsf{s}\rho_S + \mathsf{p}\rho_P + \mathsf{v}_A j^A + \mathsf{a}_A j_5^A$, two sources $w \neq w'$ with different ground states and the same densities would give $E < E' + \int (w - w')\cdot n$ and $E' < E + \int (w' - w)\cdot n$, so $E + E' < E + E'$; hence $E[w] = \min_n(F[n] + \int w\cdot n)$. **Kohn–Sham.** Varying $T_s + \int w\cdot n + E_H + E_{xc}$ with respect to $\Phi_k^\dagger$ under $\int \Phi_k^\dagger \Phi_l = \delta_{kl}$ (multipliers λ_k), with $\delta\rho_S/\delta\Phi^\dagger = \Gamma^0\Phi$, $\delta\rho_P/\delta\Phi^\dagger = \Gamma^0 i\chi\Phi$, $\delta j^A/\delta\Phi^\dagger = \Gamma^0\Gamma^A\Phi$, $\delta j_5^A/\delta\Phi^\dagger = \Gamma^0\Gamma^A\chi\Phi$, gives $H_{\mathrm{KS}}\Phi_k = \lambda_k\Phi_k$ with

$$\begin{aligned} H_{\mathrm{KS}} &= \Gamma^0\Gamma^i(-i\partial_i + \mathcal{V}_i) + \Gamma^0(m + \mathcal{S}) + \mathcal{V}_0 + \Gamma^0 i\chi\mathcal{P} + \chi\mathcal{A}_0 + \Gamma^0\Gamma^i\chi\mathcal{A}_i, \\ \mathcal{S} &= \mathsf{s} + g_S\rho_S + \frac{\delta E_{xc}}{\delta\rho_S}, \qquad \mathcal{P} = \mathsf{p} + g_P\rho_P + \frac{\delta E_{xc}}{\delta\rho_P}, \\ \mathcal{V}_A &= \mathsf{v}_A + g_V j_A + \frac{\delta E_{xc}}{\delta j^A}, \qquad \mathcal{A}_A = \mathsf{a}_A + g_A j_{5A} + \frac{\delta E_{xc}}{\delta j_5^A},\end{aligned}$$

where $E_H = \int[\frac{g_S}{2}\rho_S^2 + \frac{g_P}{2}\rho_P^2 + \frac{g_V}{2}j^A j_A + \frac{g_A}{2}j_5^A j_{5A}]$ and $T_s = \sum_k \int \Phi_k^\dagger H_{\mathrm{D}}\Phi_k$. **Decision (no sea):** the orbitals are the occupied positive-energy states. **Exchange.** Wick's theorem for a Slater determinant, $\langle c_a^\dagger c_c^\dagger c_d c_b\rangle = \varrho_{ba}\varrho_{dc} - \varrho_{da}\varrho_{bc}$ with $\varrho_{xy} = \sum_k \Phi_{k,x}\Phi_{k,y}^*$ and $\mathcal{R} = \varrho\,\Gamma^0$, gives $\langle : (\bar\Psi O_1\Psi)(\bar\Psi O_2\Psi) : \rangle = \mathrm{tr}(O_1\mathcal{R})\mathrm{tr}(O_2\mathcal{R}) - \mathrm{tr}(O_1\mathcal{R}O_2\mathcal{R})$, hence the exact exchange energy

$$E_x = -\frac{g_S}{2}\mathrm{tr}(\mathcal{R}\mathcal{R}) - \frac{g_P}{2}\mathrm{tr}(i\chi\mathcal{R}\,i\chi\mathcal{R}) - \frac{g_V}{2}\mathrm{tr}(\Gamma^A\mathcal{R}\Gamma_A\mathcal{R}) - \frac{g_A}{2}\mathrm{tr}(\Gamma^A\chi\mathcal{R}\Gamma_A\chi\mathcal{R}).$$

Chiral Kohn–Sham. With $\Phi_{kL} = E_-^\dagger\Phi_k$, $\Phi_{kR} = E_+^\dagger\Phi_k$ and the blocks above, $\kappa = \sum_k \Phi_{kL}^\dagger\Phi_{kR}$, $\rho_S = \kappa + \kappa^*$, $\rho_P = i(\kappa - \kappa^*)$, $\mathcal{M} = m + \mathcal{S} + i\mathcal{P}$, $\mathcal{V}_A^{L,R} = \mathcal{V}_A \mp \mathcal{A}_A$:

$$\begin{aligned} \big[\mathcal{V}_0^L - \tau_i^O\partial_i - i\tau_i^O\mathcal{V}_i^L\big]\Phi_L + \mathcal{M}\Phi_R &= \lambda\Phi_L, \\ \mathcal{M}^*\Phi_L + \big[\mathcal{V}_0^R + \tau_i^O\partial_i + i\tau_i^O\mathcal{V}_i^R\big]\Phi_R &= \lambda\Phi_R, \\ e^{i\vartheta\chi} &= \cos\vartheta\,1_{16} - \sin\vartheta\,\Gamma_8^{(1,7)} : \qquad \kappa \to e^{2i\vartheta}\kappa, \\ \Delta E_H &= \frac{g_S - g_P}{2}\big[(e^{4i\vartheta} - 1)\kappa^2 + (e^{-4i\vartheta} - 1)\kappa^{*2}\big].\end{aligned}$$

Uniform matter. Units $m = 1$; all 8 spin states filled to k_F; $|S^6| = 16\pi^3/15$ is the area of the unit sphere in $\mathbb{R}^7$ and $c_7 = 8|S^6|/(2\pi)^7 = 1/(15\pi^4)$. With the energy density $\mathcal{E}$, its kinetic part $\mathcal{E}_{\mathrm{kin}}$, ρ_S and the density ρ:

$$\rho = \frac{k_F^7}{105\pi^4}, \qquad \rho_S = c_7\int_0^{k_F}\frac{k^6 m^*\,dk}{\sqrt{k^2 + m^{*2}}}, \qquad \mathcal{E}_{\mathrm{kin}} = c_7\int_0^{k_F} k^6\sqrt{k^2 + m^{*2}}\,dk,$$

$$m^* = m + g_S\rho_S(m^*), \qquad \mathcal{E} = \mathcal{E}_{\mathrm{kin}} - \frac{(m^* - m)^2}{2g_S} + \frac{g_V}{2}\rho^2,$$

$$E/A = \frac{\mathcal{E}}{\rho}, \qquad E_F = g_V\rho + \sqrt{k_F^2 + m^{*2}}.$$

Because $\partial\mathcal{E}/\partial m^* = 0$ is the gap equation, $d\mathcal{E}/d\rho = E_F$ at every density, and $\frac{d}{dk_F}\frac{\mathcal{E}}{\rho} = \frac{c_7 k_F^6}{\rho}(E_F - \frac{\mathcal{E}}{\rho})$ vanishes at saturation: $E/A = E_F$ (Hugenholtz–Van Hove). **Decisions:** $g_S = -10^5$, $g_V = 10^4$, $g_P = g_A = 0$ (binding and saturation); root finding by Brent's method with 50-digit (gap equation) and 40-digit (saturation) working precision. **Dirac sea.** With a cutoff Λ_c, $g_S = g_P = g$ and a uniform $\mathcal{M}$:

$$J(M) = \frac{1}{120\pi^4}\int_0^{\Lambda_c} \frac{q^6 dq}{\sqrt{q^2+M^2}}, \qquad K(M) = \frac{1}{120\pi^4}\int_0^{\Lambda_c} q^6\sqrt{q^2+M^2}\,dq,$$

$$\kappa = -4\mathcal{M}^* J(|\mathcal{M}|), \qquad \mathcal{M} = m - 8gJ(|\mathcal{M}|)\mathcal{M}, \qquad g_c = -\frac{1}{8J(0)} = -\frac{90\pi^4}{\Lambda_c^6},$$

$$\mathcal{E}_{\rm vac} = -8K + 8J\,\mathrm{Re}[(\mathcal{M}-m)\mathcal{M}^*] + 32g|\mathcal{M}|^2 J^2.$$

Decisions: $\Lambda_c = 1$, $g = \frac{6}{5}g_c$, and $m = \frac{1}{100}$ for the explicit-breaking example.

6.4 Literature used in the second stage

- P. A. M. Dirac, Proc. R. Soc. Lond. A 117 (1928) 610 — $(i\gamma^\mu\partial_\mu - m)\psi = 0$, $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}$.
- P. Hohenberg and W. Kohn, Phys. Rev. 136 (1964) B864 — $E[v] = \min_n(F[n] + \int vn)$; W. Kohn and L. J. Sham, Phys. Rev. 140 (1965) A1133 — $[-\frac{1}{2}\nabla^2 + v + v_H + v_{xc}]\varphi_k = \varepsilon_k\varphi_k$, $n = \sum_k |\varphi_k|^2$ (atomic units).
- A. K. Rajagopal and J. Callaway, Phys. Rev. B 7 (1973) 1912; R. M. Dreizler and E. K. U. Gross, *Density Functional Theory*, Springer (1990) — the energy as a functional of the current j^A, the Dirac form of the Kohn–Sham equations, the no-sea approximation.
- G. C. Wick, Phys. Rev. 80 (1950) 268; M. Fierz, Z. Phys. 104 (1937) 553 — contractions in a Slater determinant (above), and $\sum_K (\Gamma_K)_{ab}(\Gamma_K^{-1})_{cd} = 16\delta_{ad}\delta_{cb}$ for the 256 products Γ_K of the generators.
- J. D. Walecka, Ann. Phys. 83 (1974) 491 — mean-field saturation of uniform matter, $m^* = m - \frac{g_\sigma^2}{m_\sigma^2}\rho_S$, $V_0 = \frac{g_\omega^2}{m_\omega^2}\rho$ (here $g_S = -g_\sigma^2/m_\sigma^2$, $g_V = g_\omega^2/m_\omega^2$); N. M. Hugenholtz and L. Van Hove, Physica 24 (1958) 363 — $E/A = E_F$ at saturation.
- Y. Nambu and G. Jona-Lasinio, Phys. Rev. 122 (1961) 345 — $M = m - 2G\langle\bar\psi\psi\rangle$ with a cutoff and a critical coupling (here $G = -g/2$).

6.5 Verification of the second stage

Sections 7–9 of 7+1.nb contain 74 checks, all True; they are rows of section 4.2 above, and the evaluation is the one of section 4.1. `tools/ref_dirac16.py` recomputes the constructions independently with 90 boolean checks, all True, and its output is printed in section 4.3, together with that of `tools/verify_7p1_nb.py`, which compares the printed spinors, the closed-form integrals and 23 printed numbers with it (at least 17 digits) and the first part of the notebook with the backup.

7. Authorship

Notebook, tools and documentation written by Claude (Anthropic, model claude-opus-5-5) for Nash: the first stage (sections 0–6 of 7+1.nb) from the task statement and the notebook 4+4.nb described in section 1, the second stage (sections 7–10) from the task of section 6.1; this document was generated on 2026-10-26.